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GEOGRAPHY · India: drainage system

UNDERSTAND IT FIRST

What is it? Drainage means the flow of water through **rivers and their tributaries**. The area drained by a river and its tributaries is its **basin (catchment)**; the high land that separates two basins is a **water divide**. India's rivers fall into two families: the **Himalayan rivers** (Indus, Ganga, Brahmaputra) and the **Peninsular rivers** (Godavari, Krishna, Kaveri, Mahanadi, Narmada and others). Rivers give **water, fertile soil, power and transport**, and are the basis of **inter-State water disputes**. Learn it as **drainage patterns → two families compared → Indus, Ganga, Brahmaputra → east- and west-flowing Peninsular rivers → lakes → river linking and disputes → AP**.

Drainage patterns

Point	Himalayan rivers	Peninsular rivers
Water	**Perennial**: fed by **snow, glaciers and rain**	Mostly **seasonal**: depend on **rain**
Age and course	**Young**; deep **gorges**; many are **antecedent** (older than the mountains); **meander** and shift course on the plains	**Old**, with broad, shallow valleys; fixed courses; **graded** profiles
Basins and mouths	Very large basins; big **deltas** (Ganga–Brahmaputra, the world's largest)	Smaller basins; east-flowing rivers make **deltas**, west-flowing ones **estuaries**

1. The Himalayan rivers

System	Facts
Indus	Rises in **Tibet near Lake Mansarovar**, enters India in **Ladakh**, then flows through Pakistan to the **Arabian Sea**. Tributaries (the 'five rivers' of Punjab): **Jhelum** (from **Verinag**; flows through **Wular lake**), **Chenab** (largest tributary), **Ravi**, **Beas** (from near **Rohtang**) and **Sutlej** (from **Rakas Tal**; enters India at **Shipki La**; **Bhakra Nangal** dam). The **Indus Waters Treaty (1960, World Bank)** gave the **eastern rivers (Ravi, Beas, Sutlej)** to India and the **western rivers (Indus, Jhelum, Chenab)** mainly to Pakistan; India put the treaty **'in abeyance' in April 2025**.
Ganga	The **Bhagirathi** (from **Gangotri glacier, Gaumukh**) and the **Alaknanda** meet at **Devprayag** to form the Ganga; India's **longest river** (about **2,525 km**) and **National River (2008)**. **Left-bank tributaries: Ramganga, Gomti, Ghaghara, Gandak, Kosi** (the **'sorrow of Bihar'** for its floods). **Right-bank:** the **Yamuna** (from **Yamunotri**; with **Chambal, Betwa, Ken** from the plateau) and the **Son** (from **Amarkantak**). The **Farakka barrage** feeds the **Hooghly**; the Ganga enters Bangladesh as the **Padma** and forms the **Sundarbans** delta with the Brahmaputra.

Brahmaputra	Rises at the **Chemayungdung glacier** near Mansarovar; called the **Tsangpo** in Tibet, the **Dihang (Siang)** in Arunachal, the **Brahmaputra** in Assam (after joining the **Dibang and Lohit**) and the **Jamuna** in Bangladesh. A **braided** river that carries huge silt and causes floods; **Majuli** is one of the world's largest river islands. China began a very large dam on the **Yarlung Tsangpo** (lower reaches, **2025**), raising concerns downstream.

Panch Prayag

2. The Peninsular rivers

River	Facts
Godavari	The **longest Peninsular river** (about **1,465 km**), called the **'Dakshin Ganga'**; rises at **Trimbakeshwar (Nashik)**; tributaries **Pranhita** (Wardha + Wainganga), **Indravati, Sabari, Manjira**; forms a large delta in AP; **Polavaram** project; **Papikonda** gorge.
Krishna	Rises at **Mahabaleshwar** (Maharashtra); about **1,400 km**; tributaries **Koyna, Bhima, Ghataprabha, Malaprabha, Tungabhadra, Musi**; dams **Almatti, Srisailam, Nagarjuna Sagar**, **Prakasam barrage** (Vijayawada); delta in AP.
Mahanadi, Kaveri, Pennar	**Mahanadi**: from **Sihawa** (Chhattisgarh); **Hirakud** dam (Odisha). **Kaveri**: from **Talakaveri** (Brahmagiri hills, Karnataka); gets rain from **both monsoons**, so it flows almost all year; **Shivanasamudra** falls; **KRS** and **Mettur** dams. **Pennar**: from the **Nandi hills** (Karnataka); flows through **Anantapur, Kadapa and Nellore**. **Vamsadhara** and **Nagavali** drain north coastal AP.
West-flowing	**Narmada** (from **Amarkantak**; flows west through a **rift valley** between the **Vindhyas and Satpuras**; **Dhuandhar falls** at Jabalpur; **Sardar Sarovar** dam) and **Tapi** (from **Multai**, Betul) form **estuaries**; also **Sabarmati**, **Mahi** (crosses the **Tropic of Cancer twice**), **Sharavati** (**Jog falls**), **Periyar**, **Mandovi, Zuari**; the **Luni** ends in the **Rann of Kutch**.

3. Lakes, river linking and disputes

- **Lakes: Wular** (largest freshwater lake, J&K), **Dal**; **Loktak** (Manipur; floating vegetation called **phumdis**; **Keibul Lamjao**, the only floating national park); **Chilika** and **Pulicat** (lagoons); **Kolleru** (AP); **Sambhar** (salt); **Vembanad** (Kerala; longest lake); **Hussain Sagar** (Hyderabad).
- **River linking:** the **National Perspective Plan (1980)**; the **Ken–Betwa link** (first under the national plan; foundation laid **December 2024**); AP's **Pattiseema lift** (2016) moved **Godavari water to the Krishna** through the Polavaram right canal. **Disputes: Article 262** and the **Inter-State River Water Disputes Act, 1956**; tribunals for the **Krishna** (**Bachawat**, 1970s; **Brijesh Kumar**, 2010), **Godavari** (**Bachawat**, 1980), **Vamsadhara** (AP–Odisha, award **2017**) and **Kaveri** (award 2007; Supreme Court verdict **2018**; **Cauvery Water Management Authority**).

EXAM TRAP

Basin, water divide; patterns: **dendritic (Ganga), trellis, radial (Amarkantak), centripetal (Loktak), antecedent (Indus, Sutlej, Brahmaputra)**; Himalayan **perennial, young, antecedent, deltas**; Peninsular

seasonal, old, graded; **Indus**: Mansarovar, Ladakh; Jhelum (**Verinag, Wular**), **Chenab largest**, Ravi, Beas (**Rohtang**), Sutlej (**Rakas Tal, Shipki La, Bhakra**); **IWT 1960** (east to India, west to Pakistan; **in abeyance April 2025**); **Ganga**: Bhagirathi (**Gaumukh**) + Alaknanda at **Devprayag**; **~2,525 km**; **National River 2008**; left: Ramganga, Gomti, Ghaghara, Gandak, **Kosi (sorrow of Bihar)**; right: **Yamuna** (Chambal, Betwa, Ken), **Son**; **Farakka**; **Padma**; **Sundarbans**; prayags: **Vishnu (Dhauliganga), Nand (Nandakini), Karna (Pindar), Rudra (Mandakini), Dev (Bhagirathi)**; **Brahmaputra**: Chemayungdung; **Tsangpo, Dihang (Siang), Jamuna**; braided; **Majuli**; **Godavari ~1,465 km, Dakshin Ganga, Trimbakeshwar, Pranhita, Indravati, Polavaram**; **Krishna ~1,400 km, Mahabaleshwar, Tungabhadra, Bhima, Srisailam, Nagarjuna Sagar, Prakasam barrage**; **Mahanadi (Sihawa, Hirakud)**; **Kaveri (Talakaveri; both monsoons; KRS, Mettur)**; **Pennar (Nandi hills)**; Vamsadhara, Nagavali; **Narmada (Amarkantak, rift, Dhuandhar, Sardar Sarovar)**, **Tapi (Multai)**, **Mahi (Tropic twice)**, **Sharavati (Jog)**, Luni; lakes **Wular, Loktak (phumdis, Keibul Lamjao), Chilika, Pulicat, Kolleru, Sambhar, Vembanad**; **Ken–Betwa (Dec 2024)**, **Pattiseema 2016**; **Art 262, 1956 Act**; tribunals **Krishna (Bachawat, Brijesh Kumar), Godavari (Bachawat), Vamsadhara (2017), Kaveri (2007; SC 2018; CWMA)**.

Test yourself: 5 questions (tap an option to see the answer)

PRACTICE

Consider the following statements:

1. The Indus, Sutlej and Brahmaputra are antecedent rivers.

2. The Godavari is the longest Peninsular river.

3. The Narmada and Tapi flow west through rift valleys.

4. The Chenab is a tributary of the Ganga.

Which of the above are correct?

A. 1 and 2 only

B. 2, 3 and 4 only

C. 1, 2 and 3 only

D. 1, 2, 3 and 4

Answer: C. The Chenab is the largest tributary of the Indus.

PRACTICE

Match List-I (confluence) with List-II (river that joins the Alaknanda there):

List-I (confluence)	List-II (river that joins the Alaknanda there)
A. Devprayag	**1.** Dhauliganga
B. Rudraprayag	**2.** Pindar

C. Karnaprayag	**3.** Mandakini
D. Vishnuprayag	**4.** Bhagirathi

(1) A → 4 B → 3 C → 2 D → 1

(2) A → 3 B → 4 C → 2 D → 1

(3) A → 4 B → 2 C → 3 D → 1

(4) A → 4 B → 3 C → 1 D → 2

Answer: (1) Devprayag: Bhagirathi; Rudraprayag: Mandakini; Karnaprayag: Pindar; Vishnuprayag: Dhauliganga.

PRACTICE

Which one of the following statements is NOT correct?

A. The Kosi is called the 'sorrow of Bihar'.

B. Wular is India's largest freshwater lake.

C. Loktak lake has floating vegetation called phumdis.

D. The Mahanadi rises at Mahabaleshwar.

Answer: D. The Krishna rises at Mahabaleshwar; the Mahanadi rises at Sihawa in Chhattisgarh.

PRACTICE

Assertion (A): The Godavari is often called the 'Dakshin Ganga'.

Reason (R): The Polavaram project is being built on the Godavari.

Choose the correct answer:

A. Both A and R are true and R is the correct explanation of A

B. Both A and R are true but R is not the correct explanation of A

C. A is true but R is false

D. A is false but R is true

Answer: B. Both are true, but the name comes from the river's length, size and sacredness, not from the Polavaram project.

PRACTICE

The Narmada, Son and Mahanadi all rise near Amarkantak and flow outward in different directions. What drainage pattern does this show?

A. Dendritic

B. Trellis

C. Radial

D. Centripetal

Answer: C. Rivers flowing outward in all directions from a central high point form a radial pattern.

RECALL BOX: CLOSE THE NOTES AND WRITE THESE IN YOUR OWN WORDS

1. Write the drainage patterns and the Himalayan vs Peninsular table. 2. Write the Indus, Ganga (with the prayags) and Brahmaputra systems. 3. Write the east- and west-flowing Peninsular rivers with sources and tributaries, the lakes, river linking and the water-dispute tribunals.

ECONOMY · Money supply: functions and measures

UNDERSTAND IT FIRST

What is it? Money is anything generally accepted to **buy goods and settle debts**. Before money, people used **barter** (goods for goods), which needed a **'double coincidence of wants'**: the other person had to want exactly what you had. **Money supply** is the **total stock of money held by the public** at a time. Too much money pushes up **prices (inflation)**; too little slows the economy. The **RBI** measures and manages it. Learn it as **barter problems → functions of money → kinds of money and who issues it → measures M0 to M4 → money multiplier → recent events**.

1. Functions and kinds of money

Function	Meaning
Primary	**Medium of exchange** (removes the need for double coincidence of wants) and **measure of value / unit of account** (prices of all goods stated in rupees).
Secondary	**Store of value** (savings can be kept for the future) and **standard of deferred payments** (loans repaid later in money).

Contingent	Basis of **credit**, distribution of **national income**, and a guarantee of **solvency**.

- **Kinds: commodity money** (cattle, grain), **metallic money** (coins), **paper money**, which today is **fiat money** (backed by the government's order, not by gold), and **bank (credit) money** (cheques, cards, UPI). **Legal tender** must be accepted in payment: **notes** are **unlimited** legal tender; **coins** are legal tender only **up to a limit**.
- **Who issues:** the **RBI** has the sole right to issue **banknotes** (**Section 22, RBI Act 1934**), except the **₹1 note**, which with all **coins** is issued by the **Government of India** (Ministry of Finance). Since **1956**, India follows the **Minimum Reserve System**: the RBI must keep at least **₹200 crore** in **gold and foreign exchange** (of which at least **₹115 crore** in gold) against notes issued. The **digital rupee (e₹)**, a **central bank digital currency**, was piloted from **late 2022**.

2. Measures of money supply

Measures of money supply

- These measures were set by the RBI's **Second Working Group (1977)**. **M1 (narrow money)** = currency with the public + **demand deposits** with banks + other deposits with the RBI; it is the **most liquid**. **M3 (broad money)** = M1 + **net time deposits** with banks; it is also called **aggregate monetary resources** and is the measure most watched. **M2** and **M4** add post office deposits; **M4** is the **least liquid**. (The **Reddy Working Group (1998)** suggested new measures NM1–NM3, but M1 and M3 remain the main ones.)
- **M0 (reserve money)**, also called **high-powered money**, is created by the RBI: currency in circulation + **banks' deposits with the RBI** + other deposits with the RBI. Banks lend out part of each deposit, which is deposited again and lent again, so **M3 is several times M0**. The **money multiplier** = **M3 ÷ M0**; it **falls** when the **CRR** rises or when people hold more cash. **Fisher's quantity theory: MV = PT** (money × velocity = price level × transactions): more money without more output means **higher prices**.
- **Recent events: demonetisation** of **₹500 and ₹1,000** notes on **8 November 2016** (about **86%** of currency by value); the **₹2,000** note was **withdrawn from circulation in May 2023** but **remains legal tender**; **UPI (2016, NPCI)** made India a world leader in **digital payments**.

EXAM TRAP

Barter: **double coincidence of wants**; functions: **primary (medium of exchange, measure of value)**, **secondary (store of value, deferred payments)**, **contingent (credit, income distribution, solvency)**; commodity, metallic, **fiat paper**, bank money; notes **unlimited**, coins **limited** legal tender; **RBI notes (Sec 22, RBI Act 1934)**; **₹1 note and coins: Government**; **Minimum Reserve System 1956 (₹200 cr; ₹115 cr gold)**; **e₹ pilot 2022**; **Second Working Group 1977**; **M1 = C + DD + OD (narrow, most liquid)**; **M2 = M1 + PO savings**; **M3 = M1 + net time deposits (broad, aggregate monetary resources)**; **M4 = M3 + all PO deposits except NSC (least liquid)**; **M0 = currency + bankers' deposits with RBI + OD (high-powered)**; **multiplier = M3 ÷ M0** (falls when CRR rises); **Reddy group 1998**; **MV = PT (Fisher)**; **demonetisation 8 Nov 2016 (~86%)**; **₹2,000 withdrawn May 2023 (still legal tender)**; **UPI 2016**.

Test yourself: 5 questions (tap an option to see the answer)

PRACTICE

Consider the following statements:

1. M1 includes currency with the public and demand deposits with banks.

2. M3 equals M1 plus net time deposits with banks.

3. M1 is the most liquid of the RBI's measures.

4. M3 is known as narrow money.

Which of the above are correct?

A. 1 and 2 only

B. 1, 2 and 3 only

C. 2, 3 and 4 only

D. 1, 2, 3 and 4

Answer: B. M3 is broad money; M1 is narrow money.

PRACTICE

Match List-I (function of money) with List-II (example):

List-I (function of money)	List-II (example)
A. Medium of exchange	**1.** Prices of all goods are stated in rupees
B. Store of value	**2.** Repaying a ₹10,000 loan next year
C. Standard of deferred payments	**3.** Keeping savings in money for the future
D. Measure of value	**4.** Buying vegetables with cash instead of exchanging rice

(1) A → 3 B → 4 C → 2 D → 1

(2) A → 4 B → 2 C → 3 D → 1

(3) A → 4 B → 3 C → 1 D → 2

(4) A → 4 B → 3 C → 2 D → 1

Answer: (4) Exchange: buying with cash; store: savings; deferred payments: loan repayment; measure: prices in rupees.

PRACTICE

Which one of the following statements is NOT correct?

A. ₹2,000 notes are no longer legal tender in India.

B. The RBI issues all banknotes except the ₹1 note.

C. Coins are issued by the Government of India.

D. ₹500 and ₹1,000 notes were demonetised on 8 November 2016.

Answer: A. ₹2,000 notes were withdrawn from circulation in May 2023 but remain legal tender.

PRACTICE

Assertion (A): Reserve money (M0) is called high-powered money.
Reason (R): M0 includes the time deposits of commercial banks.
Choose the correct answer:

A. Both A and R are true and R is the correct explanation of A

B. Both A and R are true but R is not the correct explanation of A

C. A is true but R is false

D. A is false but R is true

Answer: C. A is true: M0 is the base on which banks create a multiple amount of money. R is false: time deposits are part of M3, not M0.

PRACTICE

Suppose currency with the public is ₹30 lakh crore, demand deposits with banks ₹25 lakh crore, other deposits with the RBI ₹1 lakh crore and net time deposits with banks ₹150 lakh crore. What is M3?

A. ₹56 lakh crore

B. ₹206 lakh crore

C. ₹181 lakh crore

D. ₹176 lakh crore

Answer: B. M1 = 30 + 25 + 1 = 56; M3 = M1 + 150 = 206.

RECALL BOX: CLOSE THE NOTES AND WRITE THESE IN YOUR OWN WORDS

1. Write the barter problem and the functions of money. 2. Write the kinds of money, legal tender, who issues notes and coins, and the minimum reserve system. 3. Write M0 to M4 with formulas, the multiplier, Fisher's equation and recent events.

SCIENCE · Global warming, climate change, acid rain, ozone depletion, ocean acidification

UNDERSTAND IT FIRST

What is it? These are **global environmental problems** that no country can solve alone. **Global warming** is the rise in the Earth's average temperature caused by extra **greenhouse gases** from human activity; it drives wider **climate change** (heatwaves, floods, droughts, sea-level rise). **Ozone depletion** is the thinning of the **protective ozone layer** by man-made chemicals. **Acid rain** and **ocean acidification** are both caused by gases we release dissolving in water. Learn it as **greenhouse effect and gases → warming and its effects → global agreements and India → acid rain → ozone depletion → ocean acidification**.

1. Greenhouse effect and global warming

How the greenhouse effect warms the Earth

Gas	Sources and strength (global warming potential, GWP, over 100 years; CO_2 = 1)
Carbon dioxide (CO_2)	**Burning coal, oil and gas**, deforestation, cement; the **largest contributor** overall; now over **420 ppm** (about **280 ppm** before industry).
Methane (CH_4)	**Paddy fields, cattle, landfills**, coal mines, gas leaks; about **28 times** CO_2 (much more over 20 years).
Nitrous oxide (N_2O)	**Nitrogen fertilisers**, industry; about **270 times** CO_2; also **depletes ozone**.
F-gases (HFCs, PFCs, SF_6)	Refrigerants, aluminium making, **electrical switchgear**; SF_6 has the **highest GWP** (over **20,000**). Water vapour is the most abundant greenhouse gas but follows temperature.

- **How much warming:** the **IPCC (AR6)** found the Earth about **1.1°C** warmer (2011–2020) than **pre-industrial** times; **2024** was the **first calendar year above 1.5°C** (WMO). **Effects:** melting **glaciers** (Himalayan rivers at risk), **sea-level rise** (about **20 cm** since 1900), stronger **heatwaves, cyclones, floods and droughts**, **coral bleaching**, lower crop yields, spread of diseases, and climate **refugees**. Coastal AP faces **erosion** (e.g. **Uppada**) and **storm surges**.

2. Global agreements and India

Milestones of global climate action

- **IPCC (1988)**, set up by **WMO and UNEP**, assesses the science. The **UNFCCC (1992, Rio)** is the main treaty; its annual meeting is the **COP**; it rests on **'common but differentiated responsibilities and respective capabilities' (CBDR-RC)**. The **Kyoto Protocol (1997; in force 2005)** set **binding cuts only for developed (Annex I) countries**, covering **six gases**, with market tools (**CDM**, joint implementation, emissions trading). The **Paris Agreement (2015; in force 2016)** covers **all countries**: keep warming **well below 2°C** and try for **1.5°C**; each country sets its own **Nationally Determined Contribution (NDC)**, updated **every five years**; a **global stocktake** every five years (first in **2023**). The **USA** left Paris again in **2025**.
- **Recent COPs: COP26 Glasgow (2021)**: India's **Panchamrit** and **net zero by 2070**; **COP27 Sharm el-Sheikh (2022)**: a **Loss and Damage Fund**; **COP28 Dubai (2023)**: first global stocktake; call to **'transition away from fossil fuels'**; **COP29 Baku (2024):** a new finance goal of at least **$300 billion a year by 2035** for developing countries; **COP30 Belém, Brazil (November 2025)**. **India:** the **National Action Plan on Climate Change (2008)** with **eight missions** (solar, energy efficiency, sustainable habitat, water, Himalayan ecosystem, Green India, sustainable agriculture, strategic knowledge); updated **NDC (2022): 45%** cut in emissions intensity and **50% non-fossil** power capacity by 2030; **Mission LiFE**; the **Carbon Credit Trading Scheme (2023)**.

3. Acid rain, ozone depletion and ocean acidification

Problem	Cause, effects and response
Acid rain	**SO_2** (coal, smelters) and **NOx** (vehicles, power plants) form **sulphuric and nitric acids**; rain with **pH below 5.6**; harms **lakes, fish, forests, soil and marble buildings** (Taj Mahal); controlled by **flue-gas desulphurisation**, cleaner fuels and BS-VI.
Ozone depletion	**Stratospheric ozone** (15–35 km) absorbs **UV-B**; measured in **Dobson Units** (about 300 DU normal; an **ozone hole** is below **220 DU**). **CFCs**, **halons** (bromine), carbon tetrachloride and methyl bromide release **chlorine and bromine** when UV breaks them; one chlorine atom can destroy about **100,000** ozone molecules. Over **Antarctica**, **polar stratospheric clouds** speed this up, so an **ozone hole** forms each **spring (September–October)**; it was discovered in **1985**. **Effects: skin cancer, cataracts**, weaker immunity, harm to crops and **phytoplankton**. The **Montreal Protocol (1987)** has worked: the layer is expected to recover by about **2066** over Antarctica. **World Ozone Day: 16 September**.
Ocean acidification	The oceans absorb about **a quarter** of the CO_2 we emit; CO_2 + water forms **carbonic acid**, lowering pH from about **8.2 to 8.1** since industry began (about **30% more acidic**). It makes it hard for **corals, shellfish, oysters and tiny plankton** to build **calcium carbonate shells**, threatening **fisheries and reefs**; it is called the **'evil twin' of global warming**.

EXAM TRAP

Natural greenhouse effect: **~15°C vs −18°C**; GHGs **CO_2 (largest; 420+ ppm vs 280), CH_4 (~28×; paddy, cattle, landfills), N_2O (~270×; fertilisers; also ozone-depleting), HFCs, PFCs, SF_6 (highest GWP)**; water vapour; **~1.1°C (IPCC AR6)**; **2024 first year above 1.5°C**; effects (glaciers, **sea level ~20 cm**, extremes, bleaching, crops, health); AP **Uppada** erosion; **IPCC 1988 (WMO, UNEP)**; **UNFCCC 1992 (CBDR-RC)**; **Kyoto 1997/2005** (Annex I binding; 6 gases; CDM); **Paris 2015/2016** (well below 2°C, 1.5°C; **NDCs every 5 years**;

stocktake **2023**); USA out **2025**; **COP26 Glasgow (Panchamrit, 2070)**, **COP27 (Loss and Damage)**, **COP28 Dubai (transition away)**, **COP29 Baku ($300 bn by 2035)**, **COP30 Belém**; **NAPCC 2008 (8 missions)**; **NDC 2022 (45%, 50%)**; acid rain **pH < 5.6** (SO_2, NOx); ozone **15–35 km**, **Dobson Units (hole < 220 DU)**, **CFCs, halons → Cl, Br**; **1 Cl → ~1 lakh O_3**; **Antarctic spring**; discovered **1985**; UV-B effects; **Montreal 1987; recovery ~2066**; **16 September**; ocean takes **~1/4 CO_2**; pH **8.2 → 8.1 (~30%)**; shells, corals; **'evil twin'**.

Test yourself: 5 questions (tap an option to see the answer)

PRACTICE

Consider the following statements:

1. The Paris Agreement aims to keep global warming well below 2°C.

2. The Kyoto Protocol set binding emission targets for developed countries.

3. The IPCC was set up in 1988 by the WMO and UNEP.

4. The Kyoto Protocol set binding emission-reduction targets for India.

Which of the above are correct?

A. 1, 2 and 3 only

B. 1 and 2 only

C. 2, 3 and 4 only

D. 1, 2, 3 and 4

Answer: A. Under Kyoto, only developed (Annex I) countries had binding targets; India, a developing country, had none.

PRACTICE

Match List-I (greenhouse gas) with List-II (feature):

List-I (greenhouse gas)	List-II (feature)
A. Carbon dioxide	**1.** Highest global warming potential; used in electrical switchgear
B. Methane	**2.** Released from nitrogen fertilisers; also depletes ozone
C. Nitrous oxide	**3.** From paddy fields, cattle and landfills
D. Sulphur hexafluoride	**4.** Largest contributor, mainly from burning fossil fuels

(1) A → 3 B → 4 C → 2 D → 1

(2) A → 4 B → 2 C → 3 D → 1

(3) A → 4 | B → 3 | C → 2 | D → 1

(4) A → 4 | B → 3 | C → 1 | D → 2

Answer: (3) CO_2: fossil fuels; CH_4: paddy and cattle; N_2O: fertilisers; SF_6: highest GWP.

PRACTICE

Which one of the following statements is NOT correct?

A. The ozone hole is most marked over Antarctica in September–October.

B. Ozone depletion is the main cause of global warming.

C. The thickness of the ozone layer is measured in Dobson Units.

D. CFCs release chlorine that destroys ozone.

Answer: B. Global warming is caused mainly by greenhouse gases such as CO_2; ozone depletion is a separate problem that lets in more UV.

PRACTICE

Assertion (A): Ocean acidification helps coral reefs grow faster.
Reason (R): Carbon dioxide absorbed by the sea forms carbonic acid, which lowers the pH of seawater.
Choose the correct answer:

A. Both A and R are true and R is the correct explanation of A

B. Both A and R are true but R is not the correct explanation of A

C. A is true but R is false

D. A is false but R is true

Answer: D. A is false: more acidic water makes it harder for corals to build calcium carbonate skeletons. R is true.

PRACTICE

Under a global climate agreement, every country, developed or developing, sets its own emission targets and updates them every five years. Which agreement is this?

A. Kyoto Protocol

B. Paris Agreement

C. Montreal Protocol

D. Basel Convention

Answer: B. Self-set Nationally Determined Contributions, updated every five years, are the core of the Paris Agreement.

RECALL BOX: CLOSE THE NOTES AND WRITE THESE IN YOUR OWN WORDS

1. Write the greenhouse effect, the GHG table and the warming figures and effects. 2. Write IPCC, UNFCCC, Kyoto, Paris, the recent COPs, NAPCC and India's NDC. 3. Write acid rain, ozone depletion (chemistry, hole, effects, Montreal) and ocean acidification.

NOTEBOOK PAGE · Day 33: write this from memory

Close the notes. Write these headings, fill every blank, then correct in a different colour.

GEOGRAPHY: Drainage of India

Area	Write
Basics	Drainage patterns with examples; Himalayan vs Peninsular rivers
River systems	Indus (tributaries, treaty), Ganga (source, prayags, tributaries both banks), Brahmaputra (names, Majuli); east- and west-flowing peninsular rivers with source, length, tributaries, dams; lakes; river linking; disputes (Krishna, Godavari, Vamsadhara, Kaveri)

ECONOMY: Money supply

- Barter problems; functions of money (primary, secondary, contingent); kinds of money and legal tender; who issues notes and coins; minimum reserve system; M0–M4 with liquidity; high-powered money and multiplier; Fisher's equation; demonetisation, ₹2,000 withdrawal, e-rupee, UPI.

SCIENCE: Global environmental issues

Write: greenhouse effect and GHGs with GWP; warming figures and effects; IPCC, UNFCCC, Kyoto, Paris, COPs, India's NDC and NAPCC; acid rain; ozone layer, depletion chemistry, ozone hole, effects, Montreal; ocean acidification; AP's coastal risks.